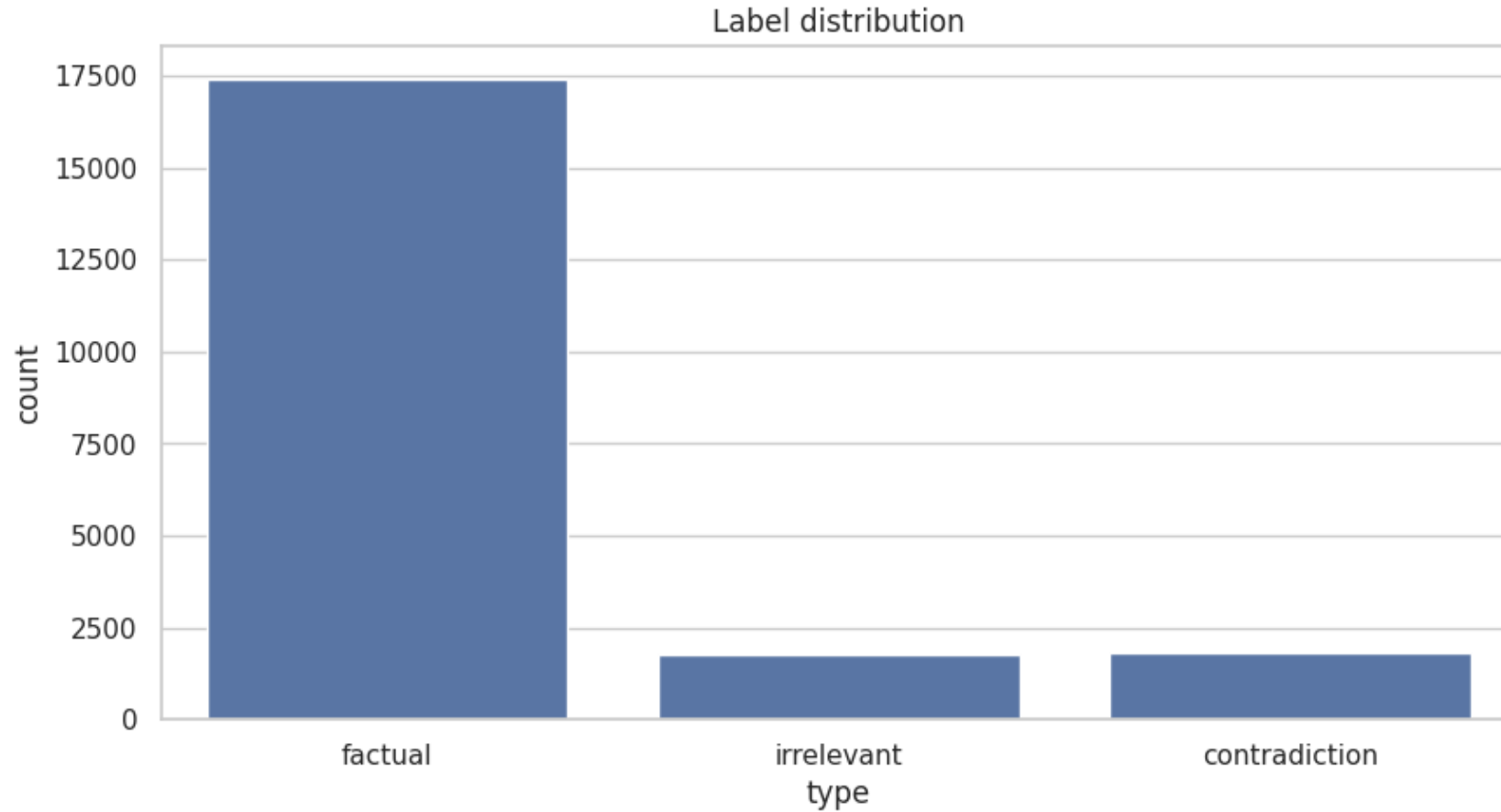


Data4Good

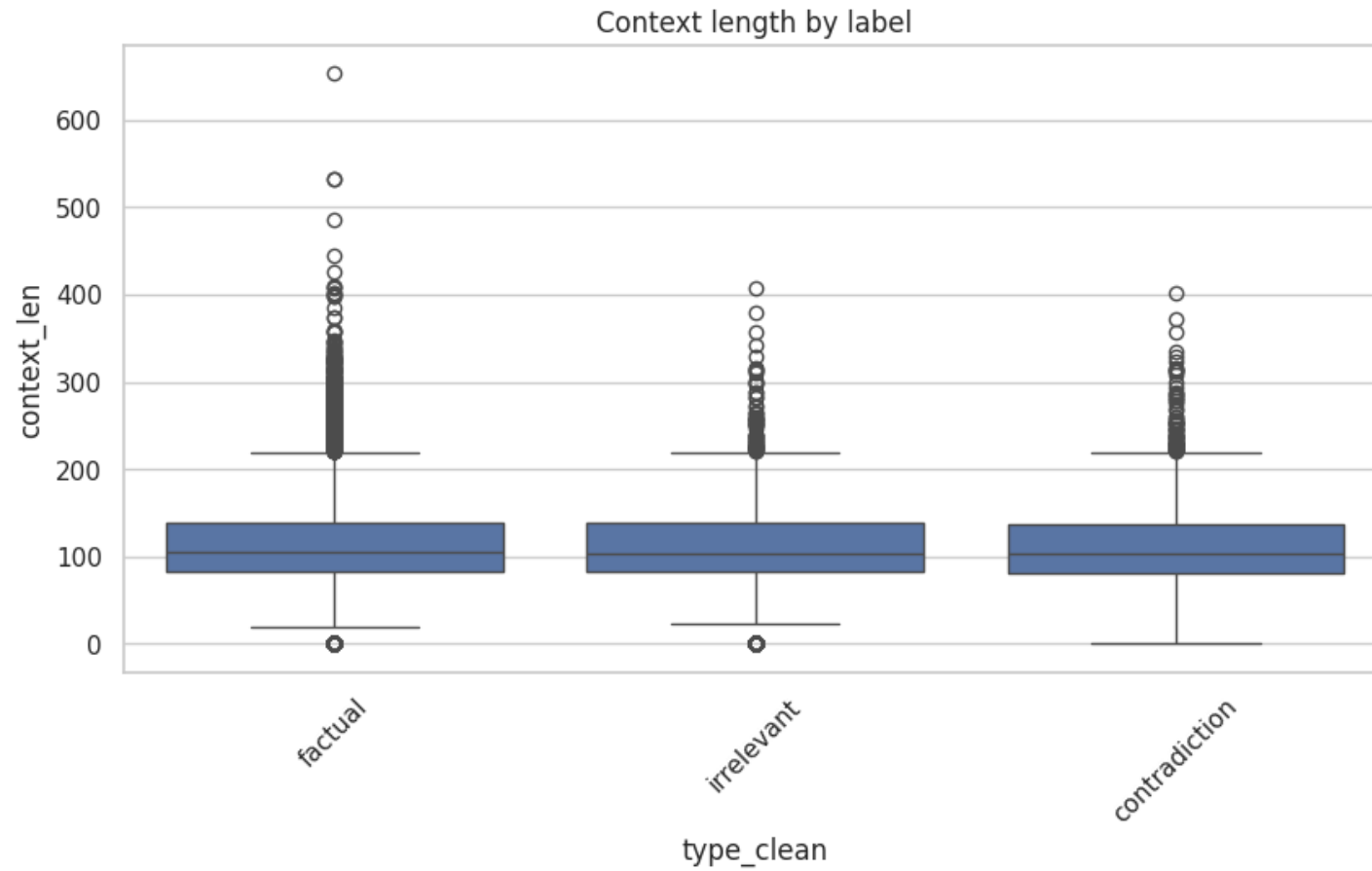
use the training set to explore, model, and classify AI-generated answers as
factual, contradiction, or irrelevant

Exploratory Data Analysis

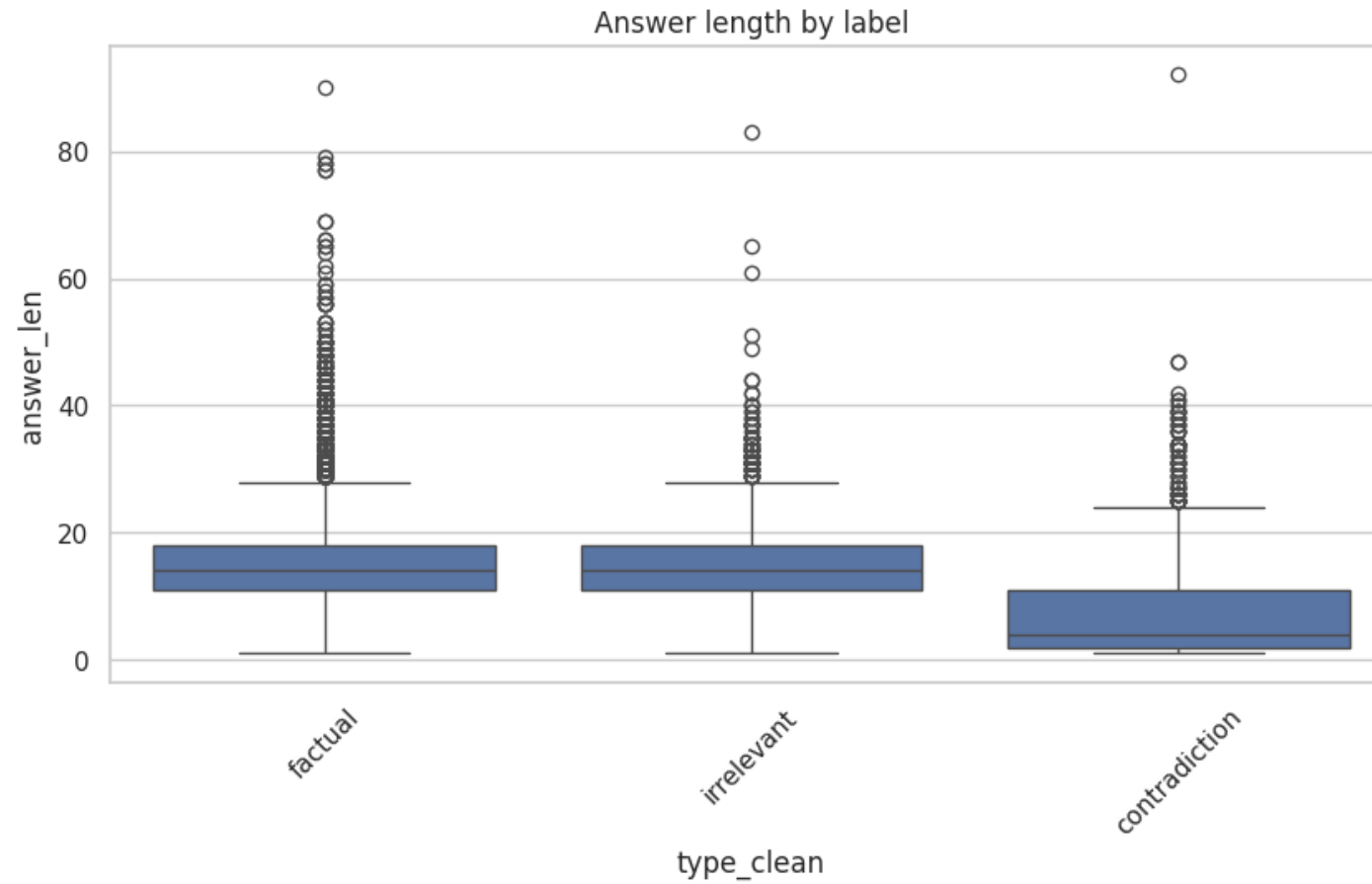
Label Distribution



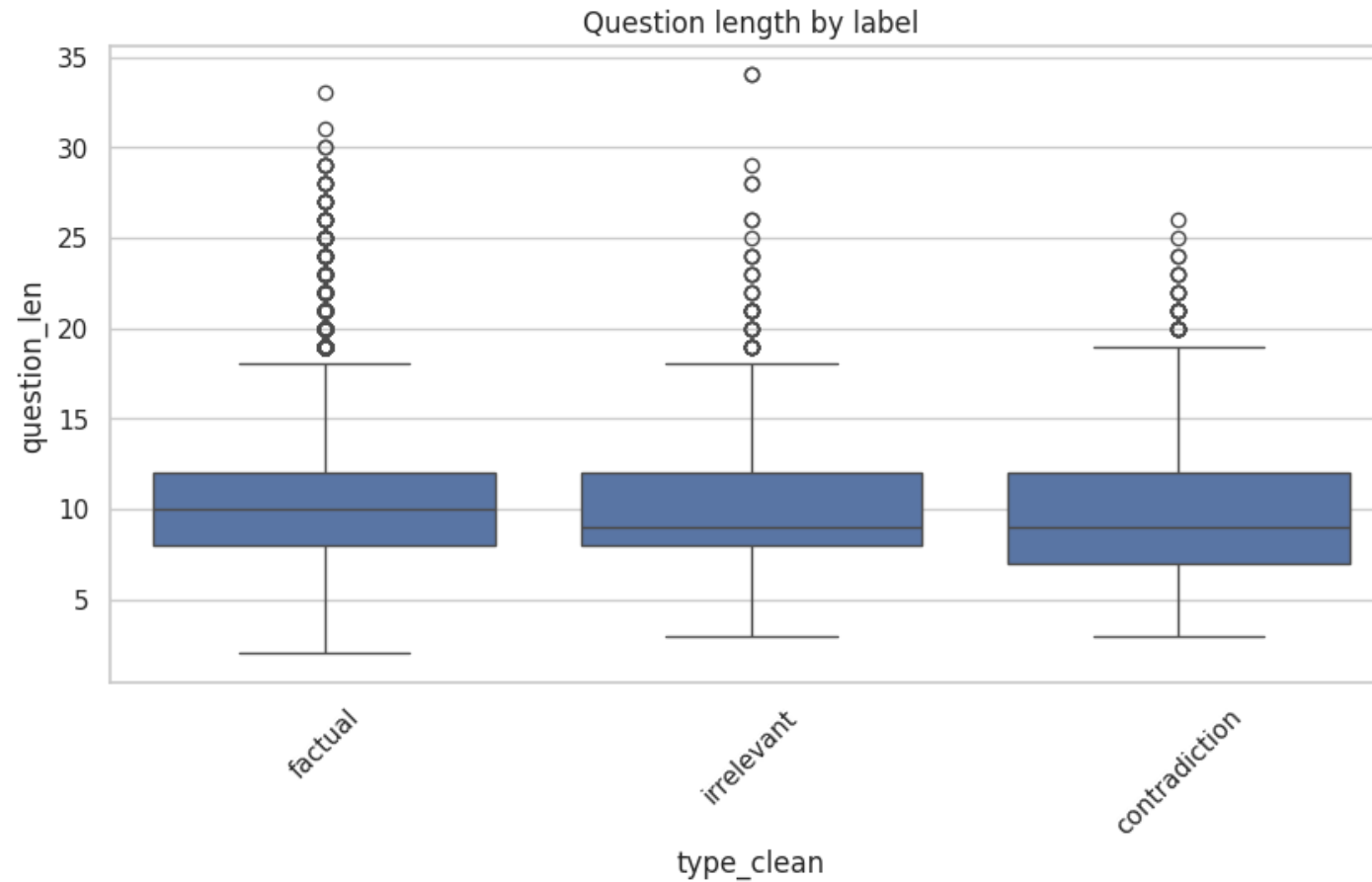
Context Length by label



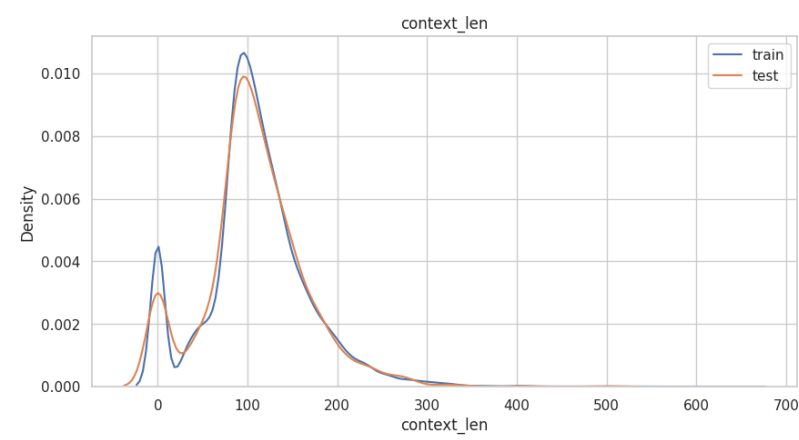
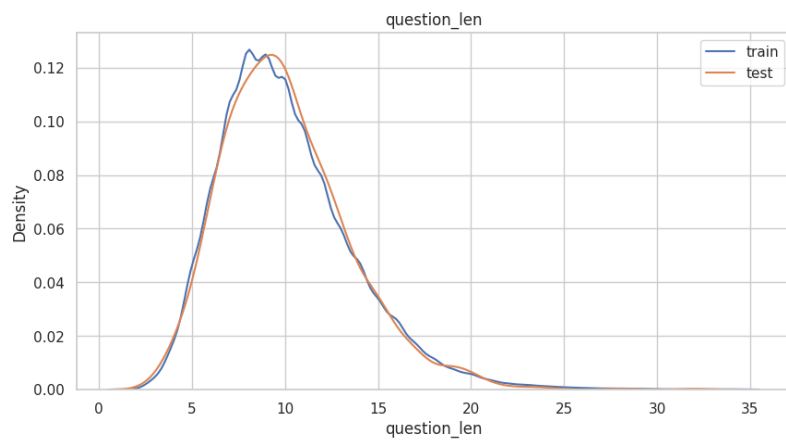
Context Length by label



Context Length by label



Test vs Train Drift



Embedding Similarity Statistics

Setup

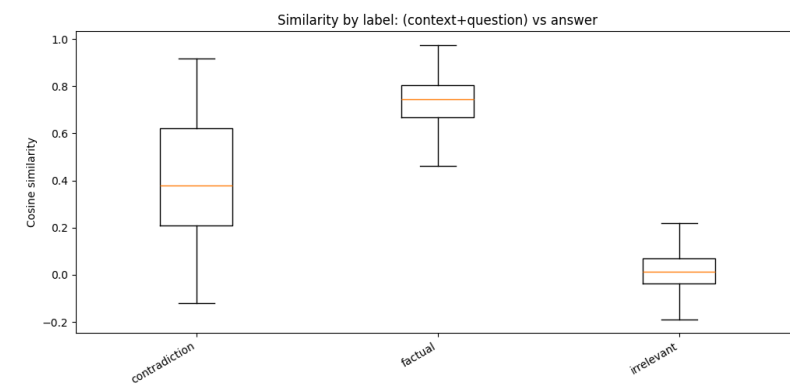
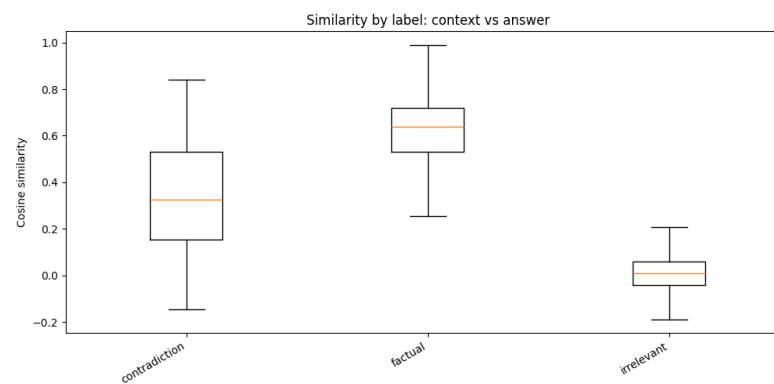
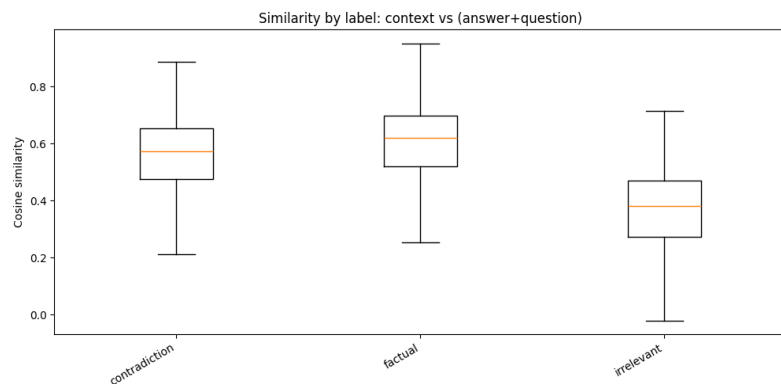
- Dataset size: **21,021 QA pairs**
- Metric: **Cosine similarity**
- Three signals:
 - Context vs (Answer + Question)
 - (Context + Question) vs Answer
 - Context vs Answer

Key Averages (Central Tendency)

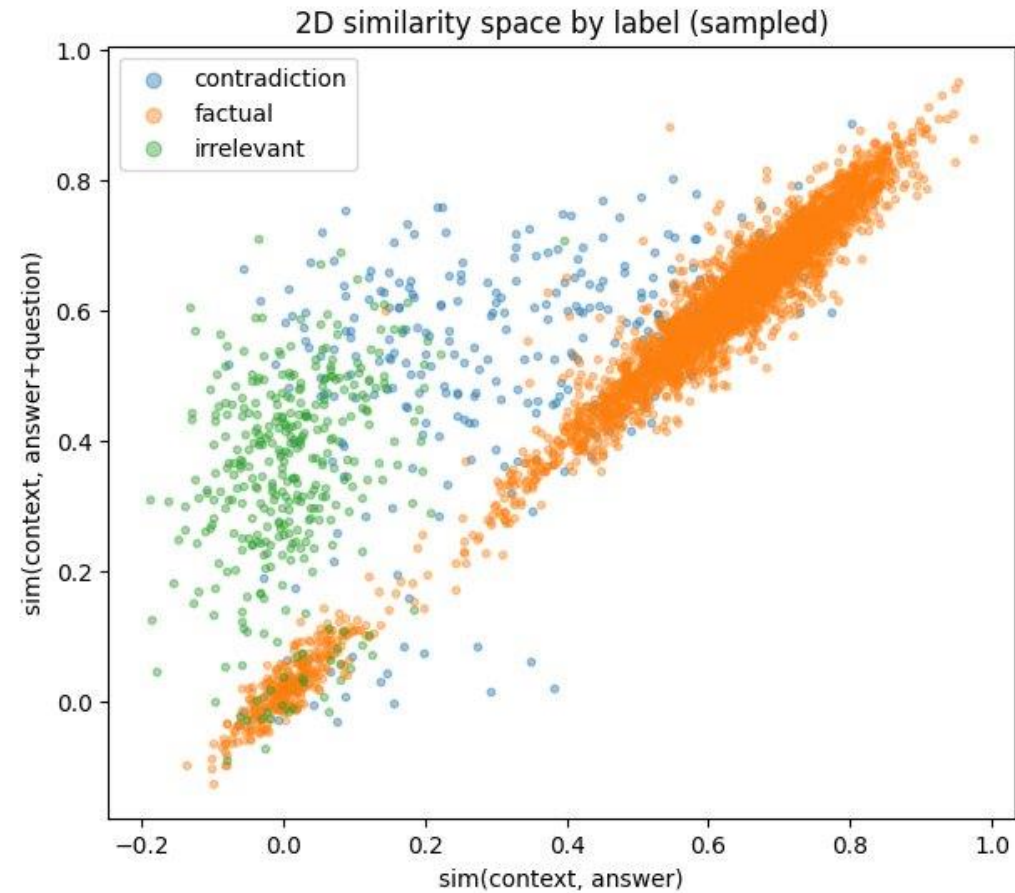
STATISTIC	SIM_CTX_VS_AQ	SIM_CQ_VS_ANS	SIM_CTX_VS_ANS
Count	21021.0	21021.0	21021.0
Mean	0.5487	0.6397	0.5160
Std	0.2067	0.2421	0.2655
Min	-0.1253	-0.2157	-0.2046
25%	0.4826	0.6021	0.4233
50%	0.6003	0.7196	0.6060
75%	0.6857	0.7926	0.7033
Max	0.9706	0.9745	0.9890

- **(Context + Question) vs Answer**
 - Mean: **0.64**
 - Median: **0.72**
 - Strongest alignment signal overall
- **Context vs (Answer + Question)**
 - Mean: **0.55**
 - Median: **0.60**
 - Moderate alignment
- **Context vs Answer**
 - Mean: **0.52**
 - Median: **0.61**
 - Weakest but still informative

By label comparison



2D Similarity Space by Label



Quartile Comparison (Robust Insight)

- Top 25% similarity:
 - CQ vs A: ≥ 0.79
 - C vs AQ: ≥ 0.69
 - C vs A: ≥ 0.70
- Bottom 25% similarity:
 - CQ vs A: ≤ 0.60
 - C vs AQ: ≤ 0.48
 - C vs A: ≤ 0.42
 - **Separation exists but is not trivial. Single thresholds will be imperfect.**

